

Wave Spawner

Introduction

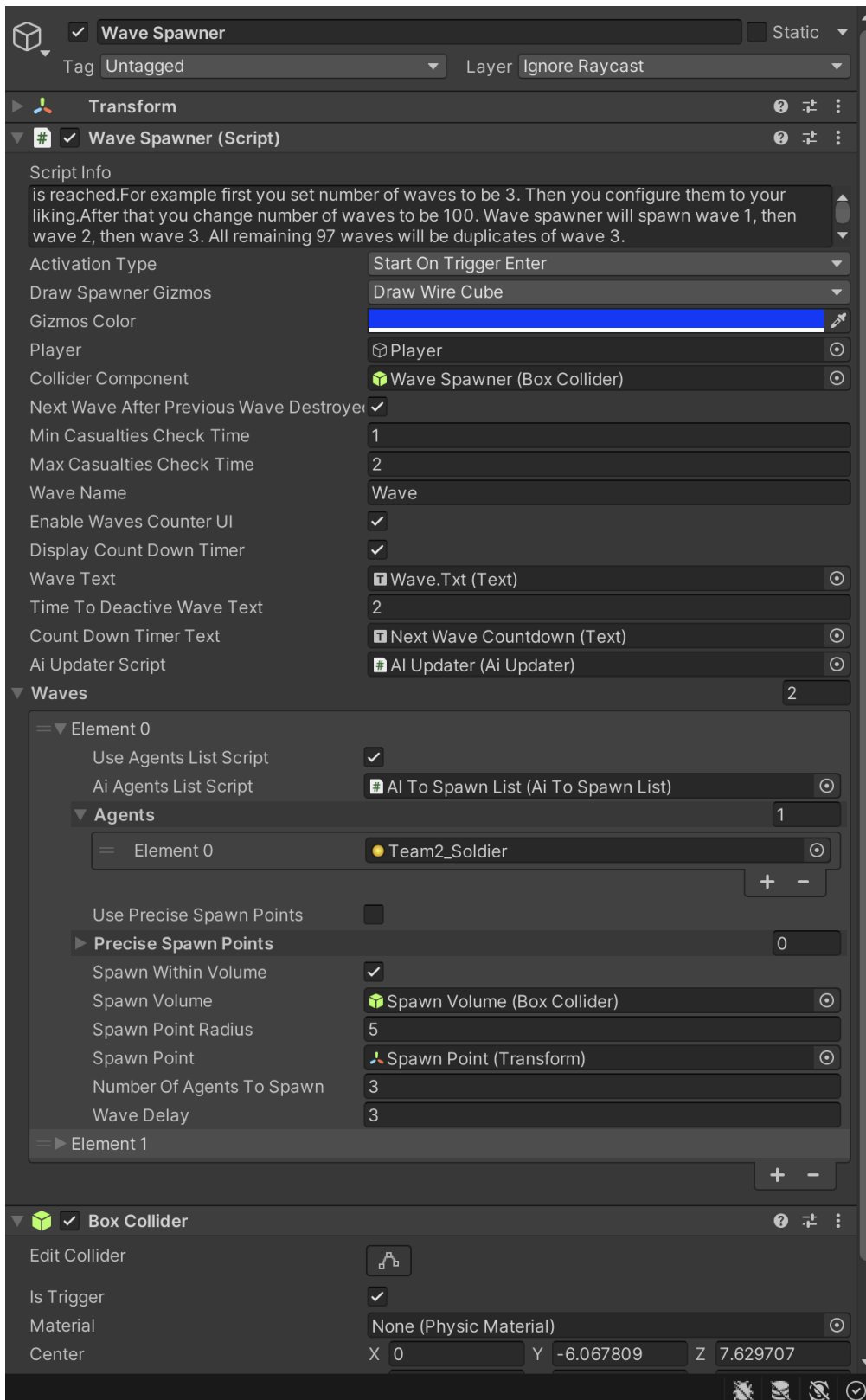
In this tutorial you will learn how to setup Wave spawner and spawn AI agents within a spawn point and collider volume. We will make sure to spawn agents on the NavMesh Surface and will spawn certain number of AI agents in each wave.

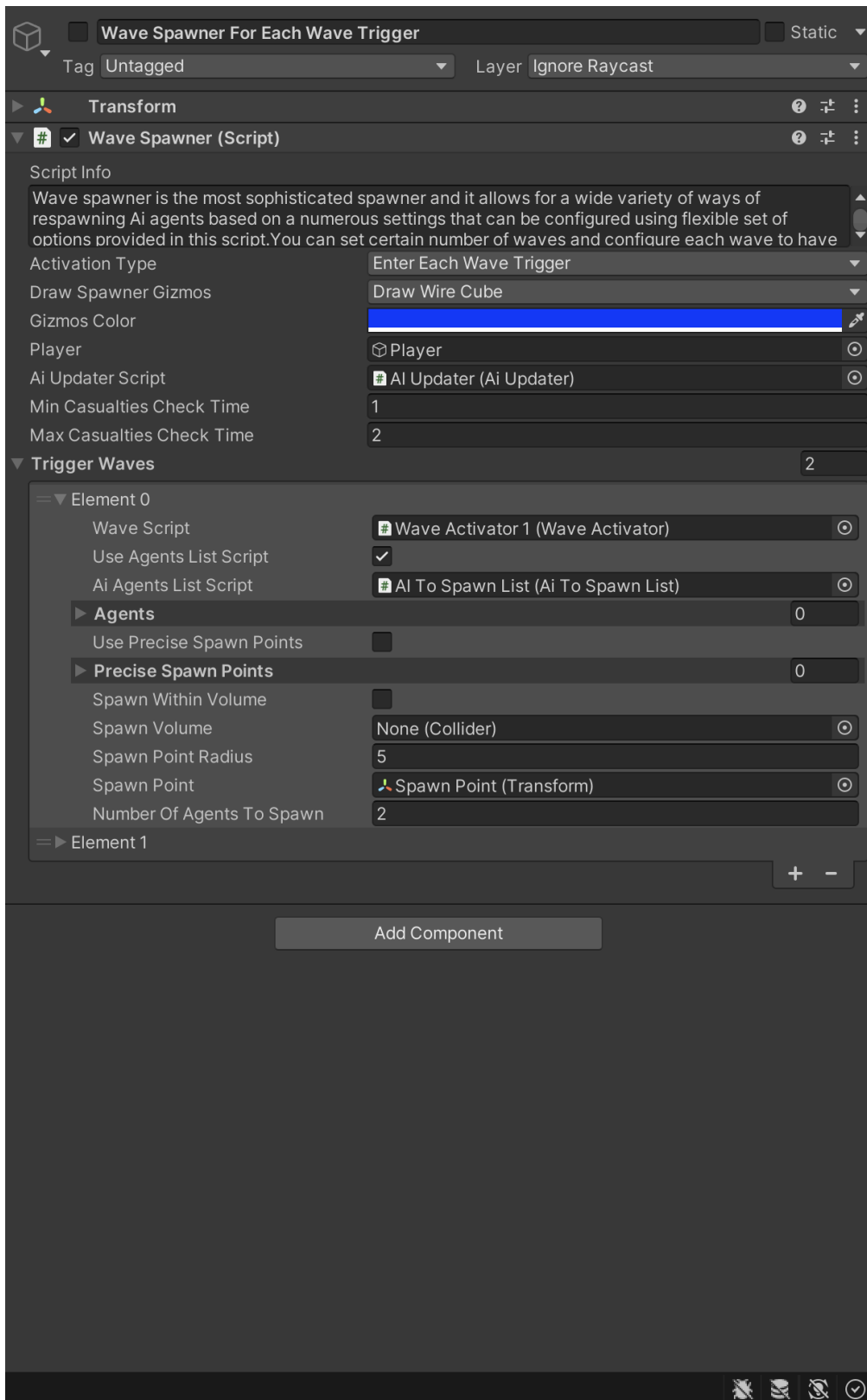
Wave Spawner

Fields	Info
ActivationType	Choose one of the three types of wave spawner activation. 'StartWaveBasedOnSingleTrigger' (StartOnTriggerEnter): Spawner begins functioning once the player enters the collider trigger attached to this gameObject with no consecutive triggering required to spawn following waves. 'StartWaveBasedOnMultipleTrigger' (EnterEachWaveTrigger): Initiates waves based on specific triggers assigned to each wave. For instance, wave 1 starts when the player enters the trigger collider for wave 1, wave 2 when player enters trigger for wave 2, and the same applies to all subsequent waves. 'ActivateAtGameStart' (BeginOnPlay): Starts functioning at the beginning of the gameplay without the need for trigger colliders.
DrawSpawnerGizmos	Draw the spawning areas (and their activation triggers) in Unity scene view to

Fields	Info
	visualize where the Ai agents will spawn in the game.
GizmosColor	Set the color of the spawning area gizmo in scene view.
Player	Drag and drop the player gameObject from the hierarchy into this field. The player gameObject will then enter this trigger area to start the spawning behavior.
ColliderComponent	Automatically assigns the Collider component to this field when the selected Wave Type option is 'StartWaveBasedOnSingleTrigger'.
NextWaveAfterPreviousWaveDestroyed	If enabled, the next wave starts only when all the agents from the previous wave are dead.
MinCasualtiesCheckTime	Minimum time between counting casualties.
MaxCasualtiesCheckTime	Maximum time between counting casualties.
maxSpawnAttemptsPerObstacle	HideInInspector.
EnableWavesCounterUI	If checked, wave counter text will be displayed in the form of '2D UI' each time a new wave starts.
WaveName	Write the word to display before showing the wave number, for example: Round, Wave, etc.
WaveText	Drag and drop the Text UI from the 2D canvas hierarchy into this field.
TimeToDeactiveWaveText	

Fields	Info
	Specify for how long the wave text will stay activated.
DisplayCountDownTimer	If checked, a countdown timer will be displayed before starting the next wave.
CountDownTimerText	Drag and drop the Text UI from the 2D canvas hierarchy into this field to display the countdown.
AiUpdaterScript	Drag and drop the Ai Updater game object from the hierarchy with the 'SpawnedAgentsTargetScriptUpdater' (AiUpdater) script attached to it into this field.
Waves	Set the total number of waves for this spawner. Each element in this list contains a set of fields and checkboxes for wave customization.
TriggerWaves	Create one or more waves that will get activated when the player hits their trigger colliders. These trigger colliders should be attached to 'WaveActivator' game objects. Waves are activated by their dedicated activators. Each wave requires such an activator, which can still be shared between different waves. To spawn the next wave, the player has to enter the trigger assigned to that wave, which stays deactivated until previous waves are destroyed.





Wave Activator

Fields	Info
TriggeringGameObject	

Fields	Info
	Drag&drop the gameobject (usually player) from the hierarchy into this field to trigger spawn event.
WaveSpawnerScript	Drag and drop the 'Wave Spawner' script attached with the parent of this gameObject into this field.
WaveStartingDelay	Specify the delay time of this wave.
DisplayCountDownTimer	If checked, the countdown timer will be displayed before starting the next wave.
CountDownTimerText	Drag and drop the Text UI from the 2D canvas hierarchy into this field to display the countdown.
DisplayWaveCounterText	If checked, wave counter text will be displayed in the form of '2D UI' that will show the number of the next wave.
TextBeforeWaveNumber	Write the message to display before showing the wave number, for example: Round or Wave.
WaveText	Drag and drop the Text UI from the 2D canvas hierarchy into this field.
WaveCounterTextDeactivationTime	Delay before deactivation of the wave-related text.

Wave Activator 1

Static

Tag UntaggedLayer Ignore Raycast

Transform

Wave Activator (Script)

Script

WaveActivator

Script Info

This script sets the various parameters of wave activation and UI displayed during this process.

Triggering Game Object

Player

Wave Spawner Script

Wave Spawner For Each Wave Trigger (Wave Spawner)

Wave Starting Delay

3

Display Count Down Timer

Count Down Timer Text

Next Wave Countdown (Text)

Display Wave Counter Text

Text Before Wave Number

Wave

Wave Text

Wave.Txt (Text)

Wave Counter Text Deactivation Time

2

Box Collider

Edit Collider

Is Trigger

Material

Center

Size

None (Physic Material)

X 0Y 0Z 0

X 5Y 5Z 5

Add Component